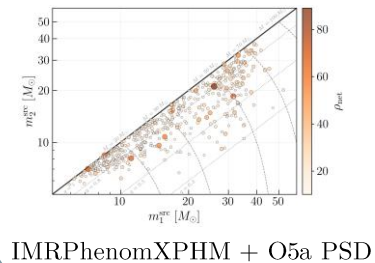


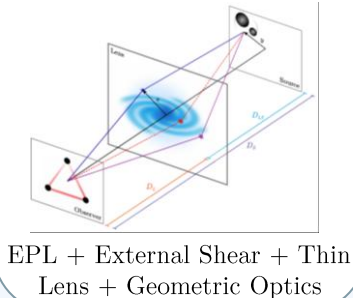
Stage 1 (Ch. 4): Population Generation

Event
Generation

Source &
Lens Population



Lensing Simulation



Detectability filter

152k candidates drawn
↓
500 detectable ($\rho_{\text{net}} \geq 10$)
↓
309 pass duty cycle (0.70)
↓
157 final events
(≥ 2 img, $\rho_{\text{net}} \geq 10$)

Stage 2 (Ch. 4): Pipeline setup

Templates

- Single-image and joint *.ini* templates + SLURM script templates

Lenskyloc_inj_hanabi_single.ini

Lenskyloc_inj_hanabi_joint.ini

File generation

Generator_inis_lenskyloc.py populates priors and injection params per event

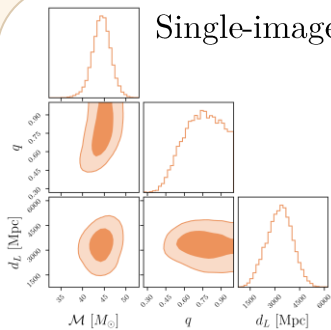
Output files

Two single-image *.ini* files and one joint *.ini* file per event + SLURM scripts for post-processing (result merging, SNR recovery, sky map generation)

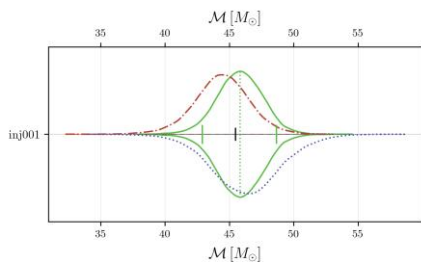
Stage 3 (Ch. 5): Parameter estimation

BILBY

Single-image PE

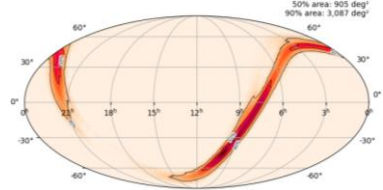


Joint PE

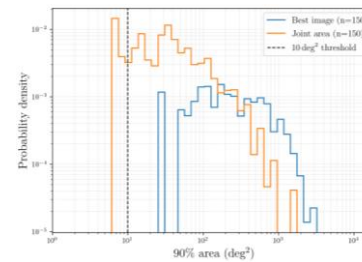


Stage 4 (Ch. 6): Sky localization

Posterior to skymap



90% credible area



Single vs joint

